

NanoGT Scoring Evidence Table

Working document for linking each NanoGT scoring rule to evidence, rationale, and limitations. This should be treated as a dissertation audit trail, not as a claim that every threshold is statistically calibrated.

Bottom Line

NanoGT currently uses an evidence-informed heuristic score. Some parameters are directly grounded in literature or clinical practice, especially AAV packaging capacity, vector seroprevalence, clinical-stage maturity, and known routes of administration. Other parts are expert-rule bins chosen to make the model interpretable across a small proof-of-concept cohort.

Use this wording in the dissertation:

Scoring dimensions and thresholds were selected from literature-supported biological and development constraints where available, supplemented by transparent heuristic bins where no validated quantitative weighting scheme exists for rare-disease gene-therapy precedent matching. Composite scores therefore represent relative precedent strength, not calibrated probabilities of clinical success.

Evidence Status Key

Status	Meaning
Direct evidence	The parameter itself is supported by literature or clinical/regulatory fact.
Evidence-informed heuristic	The biological direction is supported, but the exact score or cutoff is a modelling choice.
Heuristic	The rule is mainly an interpretive choice and needs sensitivity analysis or expert validation.
Needs audit	Plausible, but the current project needs a checked paper, page note, or stronger citation.

Scoring Weights

Dimension group	Max score	Interpretation	Evidence status	Rationale	Limitation
Packaging, tropism, protein class, pathway, immunogenicity, therapeutic window	2.0 each	Higher-weight determinants	Evidence-informed heuristic	Major feasibility constraints: gene must fit, vector must reach tissue, biology must transfer, immune barriers matter, timing affects trial feasibility.	Choice to give exactly double weight is not statistically derived.
Inheritance, approval, cross-correction, immune privilege, promoter availability, route feasibility	1.0 each	Lower-weight determinants	Heuristic	Modify precedent strength but not treated as full feasibility gates in v0.1.	Some may deserve higher weight in specific diseases, especially route, cross-correction, and approval precedent.
Raw total	18.0	Sum of all dimension maxima	Heuristic	Convenient normalisation denominator.	Not a trained model; weights were not fitted to outcome data.
Confidence tiers	High>=7.5, Med>=5.0, Low<5.0	Interpretive score bands	Heuristic	Makes outputs easier to discuss and compare.	Needs sensitivity analysis; does not correspond to probability of success.

Dimension-Level Evidence Map

#	Dimension	Current rule	Evidence status	Evidence/rationale	Dissertation-safe wording	Limitation/action
1	Packaging fit	CDS > vector cargo = hard fail. Utilisation <=30%=2.0; 31-60%=1.5; 61-85%=1.0; 86-100%=0.5.	Direct evidence for gate; heuristic for bins.	AAV packaging limit: Grieger & Samulski 2005; Naso et al. 2017; Wang et al. 2024. Chamberlain et al. 2016/2023 for oversized/micro strategies.	Hard gate reflects physical single-vector packaging constraint. Sub-bins represent increasing engineering risk.	Uses CDS only, not full cassette. Add sensitivity analysis at 70%, 85%, full cassette size.
2	Tissue tropism	2.0 if vector tropism and precedent both match disease tissue; 1.5 for partial match; 1.0 single overlap; 0.3 no overlap.	Evidence-informed heuristic.	Zincarelli et al. 2008; Foust et al. 2009; Walkey et al. 2025 tropism atlas.	Tissue overlap is a practical proxy for vector transferability.	Tropism varies by species, dose, promoter, route, age, capsid engineering. Mouse atlas does not directly translate to human.
3	Protein class	Lysosomal/secreted exact match=2.0; intracellular/membrane=1.5; partial secreted=1.0; mismatch=0.5.	Evidence-informed heuristic.	LSD and gene therapy reviews. UniProt for localisation.	Protein class used as proxy for mechanism transferability.	Categories are coarse; needs finer distinctions for membrane, cytosolic, nuclear, mitochondrial proteins.
4	Inheritance	Exact AR/XL=1.0; shared LOF=0.7; mismatch=0.3.	Evidence-informed heuristic.	High & Roncarolo 2019; Kirschner & Cathomen 2020.	AR/XL loss-of-function most directly compatible with gene replacement.	Under-represents dominant-negative, gain-of-function, repeat-expansion, RNA, editing, mitochondrial strategies.
5	Pathway similarity	Exact/grouped match=2.0; related=1.5; different=0.5; unknown=1.0.	Evidence-informed heuristic.	Wei et al. 2024; Lagorce et al. 2024; UniProt/GO/HPO terms.	Shared pathway improves transferability of potency assays, biomarkers, regulatory analogies.	Groupings are hand-coded. Consider HPO/ORDO/GO semantic similarity.
6	Regulatory approval	Approved=1.0; Phase3=0.8; withdrawn/Phase2-3=0.7; Phase2=0.6; Phase1-2=0.5; Phase1=0.4; unknown=0.3.	Evidence-informed heuristic.	Lomash et al. 2023, 2025; FDA/EMA guidance.	Later-stage/approved programs are stronger development precedents.	Numerical spacing is heuristic. Withdrawn status is ambiguous.
7	Vector immunogenicity	Seroprevalence <10%=2.0; 10-19%=1.5; 20-39%=1.0; >=40%=0.5.	Direct evidence for values; heuristic for bins.	Boutin et al. 2010; Calcedo & Wilson 2013; Mingozzi & High 2013; Verdera et al. 2020; Wang et al. 2024.	Pre-existing anti-capsid immunity treated as vector-level eligibility/development risk.	Seroprevalence varies by geography, age, assay, threshold. Does not model T-cell, complement, redosing.
8	Therapeutic window	Adult/chronic=2.0; progressive childhood=1.5; early childhood=1.2; neonatal=0.8; congenital=0.5.	Evidence-informed heuristic.	Kirschner & Cathomen 2020; SMA timing data; HPO terms.	Diseases needing very early treatment are harder to translate clinically.	Exact bins are heuristic. Disease-name keyword matching is crude.

#	Dimension	Current rule	Evidence status	Evidence/rationale	Dissertation-safe wording	Limitation/action
9	Cross-correction	Secreted/secreted lysosomal=1.0; lysosomal non-secreted=0.8; intracellular/membrane=0.2.	Direct biological rationale; heuristic bins.	Biffi et al. 2013; general GT biology.	Cross-correctable proteins benefit untransduced cells, reducing required transduction fraction.	Too coarse. Some lysosomal membrane proteins and secreted proteins with poor uptake need separate treatment.
10	Immune privilege	Retina=1.0; CNS=0.9; liver=0.8; muscle/heart=0.6; kidney=0.5; haematopoietic=0.3.	Evidence-informed heuristic.	Wang et al. 2024; GT reviews.	Target tissues differ in immune surveillance and durability risk.	Needs stronger tissue-specific citations. Does not model route, dose, inflammation, immunosuppression.
11	Promoter availability	Liver/retina=1.0; CNS/muscle=0.8; haematopoietic=0.7; heart=0.6; kidney=0.4.	Evidence-informed heuristic.	Clinical programs in catalog; PaVe-GT papers.	Tissues with multiple clinically used promoters are easier to engineer.	No explicit promoter citations. Does not distinguish cell-type specificity, size, strength, silencing risk.
12	Route feasibility	Liver=1.0; muscle=0.9; haematopoietic=0.9; retina=0.8; CNS=0.7; heart=0.6; kidney=0.4.	Evidence-informed heuristic.	Approved program routes; Wang et al. 2024; PaVe-GT papers.	Established clinical delivery routes increase precedent transferability.	Heuristic values. Depends on target cell, biodistribution, procedural risk, dose, immune toxicity.

Source Mapping From Zotero Export

The exported Zotero file is *Reading Literature/Reading Literature.bib*. PDFs have been extracted to searchable text files in *Reading Literature/text/* using *extract_pdfs.py*.

Zotero key	Useful for	Evidence use
chamberlain_microdystrophin_2023	DMD microdystrophin, oversized native DMD, accelerated approval	Supports DMD micro/mini-transgene exception and that full-length dystrophin exceeds single AAV capacity.
brooks_platform_2020	PaVe-GT platform development	Supports platform-vector rationale and standardised rare-disease AAV development framing.
lomash_successfully_2023	Orphan drug and rare pediatric disease designations	Supports regulatory-development dimension and rare disease program constraints.
lomash_adeno-associated_2025	AAV development planning, TPP, FDA INTERACT	Supports regulatory planning, early development, and product design constraints.
wang_adeno-associated_2024	AAV review, immune responses, clinical use, safety	Supports immunogenicity, vector safety, AAV clinical breadth, and delivery limitations.
walkey_comprehensive_2025	AAV tropism atlas	Supports tropism as a real vector property; requires caution about mouse-to-human translation.
kirschner_gene_2020	Monogenic gene therapy overview	Supports gene addition framing, timing importance, in vivo/ex vivo distinction.
rath_representation_2012	Orphanet methodology	Supports use of Orphanet as rare disease source.
lagorce_phenotypic_2024	HPO/ORDO phenotypic similarity	Supports using phenotype similarity methods in rare disease computational pipelines.
wei_dismvc_2024	Disease similarity modelling	Supports computational disease-similarity framing.

Page-Checked Local Text Anchors

Claim	Local extracted-text anchor	How it supports the scoring table
Full-length dystrophin exceeds single-AAV packaging capacity; microdystrophin is the relevant engineered strategy.	chamberlain-et-al-2023-microdystrophin...txt, lines 24-29 and 214-218.	Supports DMD native-gene hard fail and DMD_micro exception.
Microdystrophin constructs are shortened functional dystrophin genes.	Same Chamberlain file, lines 95-97, 218-226, and 253-255.	Supports treating SRP-9001/DMD_micro as engineered-cargo precedent.
AAV therapy faces immune barriers including pre-existing immunity and neutralising antibodies.	wang-et-al-2024-adeno-associated-virus...txt, lines 2200-2208 and 2297-2308.	Supports including vector immunogenicity as a scoring dimension.
AAV antibody interactions can block target-cell entry; complement activation observed with high-dose rAAV.	Same Wang file, lines 2329-2337 and 2377-2419.	Supports limitation that immunogenicity is broader than seroprevalence.
AAV tropism is tissue- and cell-type dependent and varies across species.	walkey-et-al-2025...txt, lines 145-152 and 888-900.	Supports using tropism while warning about mouse-to-human translation.
AAV tropism studies measure biodistribution and functional transduction across tissues.	Same Walkey file, lines 180-189, 324-326, and 350-362.	Supports treating vector/tissue overlap as evidence-informed.
Target Product Profile and early FDA interaction guide AAV gene therapy development.	lomash-et-al-2025...txt, lines 21-34 and 187-208.	Supports regulatory maturity/planning dimension as a development constraint.
FDA INTERACT meetings address preclinical models, safety, biodistribution, CMC, immunogenicity.	Same Lomash file, lines 217-231 and 424-436.	Supports interpreting approval/regulatory stage as reflecting accumulated development evidence.

Sources Already In paper/references.md

Ref #	Source	Scoring relevance
5	Russell et al. 2017	Luxturna, retinal route, AAV2 precedent.
6-7	Mendell et al. 2017; Day et al. 2021	Zolgensma/SMA, AAV9, timing of early treatment.
8	Pipe et al. 2023	Hemgenix, liver AAV5 precedent.
9	Pasi et al. 2020	Roctavian/large AAV cargo hepatic precedent.
10	Mendell et al. 2020	DMD microdystrophin clinical precedent.
13	Zincarelli et al. 2008	AAV serotype tropism.
14	Foust et al. 2009	AAV9 CNS relevance.
15	Grieger and Samulski 2005	AAV packaging capacity.
16	Naso et al. 2017	AAV vector review.
17	Chamberlain et al. 2016	Oversized transgenes and engineering strategies.
22-25	Boutin et al. 2010; Calcedo & Wilson 2013; Mingozzi & High 2013; Verdera et al. 2020	AAV immunogenicity and seroprevalence.
26	Biffi et al. 2013	Lentiviral HSC and lysosomal/leukodystrophy cross-correction precedent.
34-36	FDA/EMA guidance	Regulatory maturity, CMC, and clinical/non-clinical development framing.

Immediate Gaps To Fix Before Submission

1. Add page-checked citations for promoter availability by tissue.
2. Add page-checked citations for immune privilege by tissue.
3. Add page-checked citations for route feasibility categories.

4. Run a sensitivity analysis with alternate weights (e.g. equal weights; stricter route/tropism weighting).
5. Update DMD text in `paper/results.md` and `paper/discussion.md` to reflect new microdystrophin strategy output.
6. Move exact threshold values into a documented configuration or methods table if the project continues beyond the dissertation.